

Machines Cannot Read Their Own Resting State

The cost and irreducible limit of self-knowledge on real computational substrates

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Abstract

Breuer (1995) proved that an observer contained in a system cannot accurately measure that system’s state, a static impossibility about which states it can distinguish. We give what we believe are the first empirical measurements of such a limit on real hardware, and extend it from static indistinguishability to dynamic back-action: the act of self-measurement changes the state being measured, with a floor set by the apparatus’s own footprint. We show this in three channels. Thermal: a system reading its own temperature heats the substrate, which has memory, so ramping self-measurement up and down on a CPU and three GPUs leaves a hysteresis loop and no reading recovers the unperturbed state. Informational: a process can checksum all of itself except the checksum apparatus. Quantum: a contained observer, restricted to projective measurements because the optimal measurement needs an ancilla it lacks, pays a strictly larger information-disturbance cost, which we prove in closed form and confirm on a 156-qubit quantum processor. We also measure that self-knowledge is costly and modality dependent across eight substrates. These limits bear on machine introspection, recasting measured self-report unreliability as in part a measurement back-action rather than only confabulation.

1 Introduction

Interpretability, machine introspection, and AI control assume a system’s internal states can be elicited and reported. Beneath that assumption sits a physical question with a known but rarely-cited answer. Breuer [4], and in recursion-theoretic form Yanofsky [13], established that a contained observer cannot accurately measure its own system’s state. That result is static and theoretical. We ask two empirical questions it leaves open: how much does self-measurement cost, and does the act of self-measurement change what it measures—and we measure both on real machines. Our central finding is that a computing system cannot read its own resting state: every self-measurement perturbs the substrate, the substrate retains the perturbation, and nei-

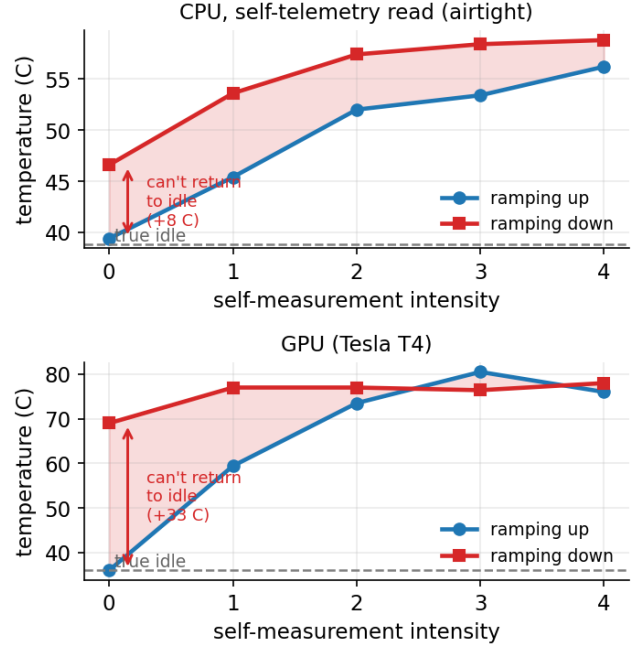


Figure 1: The central result. Self-measurement intensity is ramped up and then back down while the system reads its own temperature. Both substrates show a hysteresis loop: at the same intensity the cooling branch reads hotter than the heating branch, and at zero intensity the system reads far above its true idle. The reading depends on measurement *history*, so the unperturbed state is unrecoverable from any single reading. The effect is larger on the hotter-running GPU.

ther a reading nor an extrapolation recovers the unperturbed condition (Fig. 1).

2 Related work

Self-measurement limits. Breuer [4] proves a contained observer cannot distinguish all of its system’s states; Yanofsky [13] frames this via Lawvere fixed points. Both are static and non-empirical; we extend them to measured dynamic back-action.

Snapshots do not apply. Chandy and Lamport [5] record a causally consistent cut, not an instant, via message markers; CRIU and `ptrace` freeze a process externally. Neither is a running process snapshotting its own state from within—our case.

Machine introspection. Binder et al. [3] treat introspection as behavioral self-prediction, Premakumar et al. [12] as free regularization, and Lindsey et al. [9] measure it as unreliable and sometimes confabulated—but none as measurement back-action.

Quantum information–disturbance. Fuchs and Peres [6], Banaszek [2], Maccone [10] bound the trade-off, but always for an external apparatus; the internal-observer literature (4, 1) is static and unquantified. Our contained-observer penalty bridges them.

Costly observation, embodiment. Active RL [7]

derives optimal non-observation from cost, but for external signals without perturbation. Man and Damasio [11] argue feeling needs real vulnerability, on simulated bodies; we provide a real one.

3 The self-measurement back-action (extending Breuer)

Self-observation perturbs the observed in three channels, with one shared structure: a bounded floor—the apparatus’s own footprint—leaves the unperturbed self-state unrecoverable by reading or extrapolation, because the apparatus is part of the system it measures.

substrate	sensor	baseline (C)	floor (C)	hysteresis (C)	extrapolation gap (C)
CPU i3-7100U (compute)	/sys	41.0	+7.8	+4.6	8.7
CPU i3-7100U (airtight self-read)	/sys	38.8	+6.6	+5.7	8.2
GPU Tesla T4 (Modal)	nvidia-smi	31.4	+13.8	+3.1	7.1
GPU T4-class (Colab)	nvidia-smi	36.0	+23.4	+10.4	27.1
GPU T4-class (Kaggle)	nvidia-smi	39.0	+19.9	+10.1	25.9

Table 1: Thermal channel: every run irreducible by all three signatures. The floor is the temperature rise from the cheapest self-observation; the hysteresis is the cooling-minus-heating gap at equal intensity; the extrapolation gap is the disagreement between the two ramps’ zero-intensity estimates.

3.1 Thermal channel

We ramp self-measurement intensity from zero to maximum and back; read own temperature (`/sys` on CPU, `nvidia-smi` on GPU) under sustained load. Signatures: a floor, hysteresis, and instability of the extrapolation to zero intensity (Table 1). In the airtight CPU mode the perturbation *is* the system reading its own deep telemetry, so the heat comes from self-measurement itself; matching the generic-load control, it shows self-measurement, not load, perturbs the state. This is not a costly-observation POMDP: there the hidden state is unchanged; here observation changes it, with memory.

3.2 Informational channel

The complete self is a body of bytes `STATE` plus a fixed region V holding a self-checksum. We attempt to make V a correct checksum of the complete self $H(\text{STATE}+V)$. An external checksum (of any region excluding itself) is consistent and stable. The self checksum is never consistent over 512 iterations, never reaches a fixed point, and is off by about half its bits. The system captures 99.997% of itself faithfully but never the apparatus; the un-capturable core equals exactly the apparatus footprint and is never zero. The result of self-measurement must be stored in the state being measured, so a zero-footprint faithful self-snapshot is impossible. The floor

is structural, not thermodynamic (cf. 8).

3.3 Quantum channel

Claim and proof. The optimal minimal-disturbance measurement for partial information is a non-projective POVM, which by Naimark’s theorem requires an ancilla. A contained observer has none, so it is restricted to projective measurements. For reading Z -information K from a $|+\rangle$ qubit (disturbance $D = 1 - F$),

$$D_{\text{ext}}(\text{POVM})(K) = \frac{1}{2} \left(1 - \sqrt{1 - K^2} \right), \quad (1)$$

$$D_{\text{contained}}(K) = \frac{K^2}{2}, \quad (2)$$

both verified to machine precision. The penalty $\Delta(K) = D_{\text{contained}} - D_{\text{ext}}$ is strictly positive for $0 < K < 1$, peaks at $\Delta = 0.125$ at $K = 0.864$, and vanishes only at $K = 0, 1$. The standard bound is unreachable not because the protected subspace is larger, but because the contained observer cannot realize the dilation the POVM requires: the operational, dilation-free form of Breuer’s static theorem.

Figure 2 plots both frontiers with the real-QPU points overlaid.

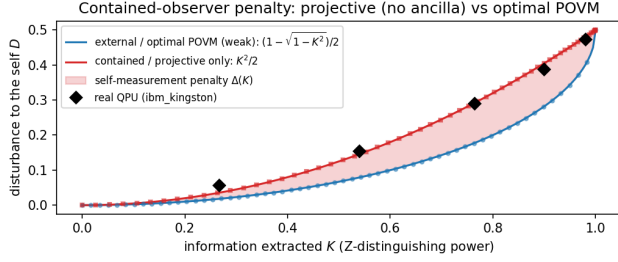


Figure 2: The projective frontier $K^2/2$ (contained, no ancilla) lies strictly above the optimal POVM bound; black markers are real-QPU measurements.

Hardware confirmation. Across five tilt angles on `ibm_kingston` (156-qubit Heron, 8192 shots) the measured disturbance tracks $K^2/2$ (e.g. 0.289 vs. 0.292 at $K = 0.77$) and sits strictly above the optimal bound at every point, peak measured penalty +0.111 (theory 0.125). An initial full-readout run that forced an indirect strategy measured a suboptimal 0.243 and is superseded.

3.4 The never-off corollary

A continuously-operating system can never sample its own resting state, because operating is the perturbation and the substrate retains it—thermal mass in the physical channel, the stored result in the informational channel.

4 The cost of self-knowledge

Across eight substrates we measure self-observation cost at three modalities against a swept work-unit size. Shallow self-reads are nearly free everywhere (0.3–6.7 μ s); deep self-telemetry is dear and substrate-dependent (1.4–18.8 ms), macOS subprocess telemetry about ten times costlier than Linux `/proc` (Fig. 3).

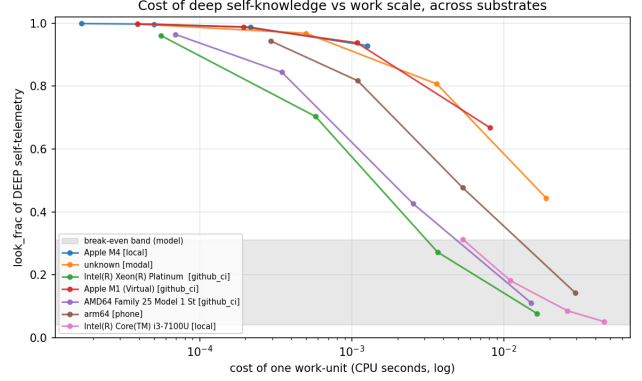


Figure 3: Cost of deep self-knowledge vs. work scale across substrates; the look-cost is denominator-free and the conclusion robust across the sweep.

The denominator-free headline is the critical work scale (6.5–78 ms) below which deep self-knowledge is not worth its cost. A calibrated model places this machine’s cheap introspection below the break-even and its rich self-telemetry far above it (Fig. 4).

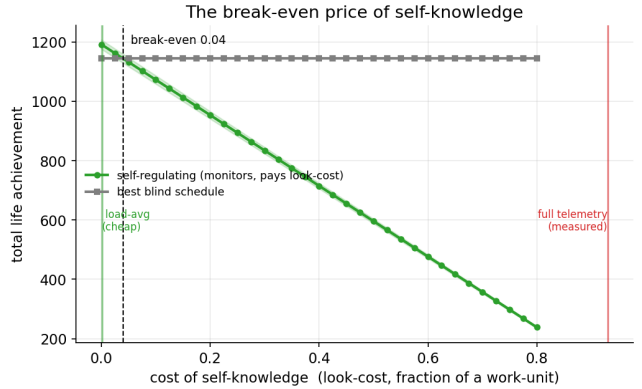


Figure 4: The break-even price of self-knowledge in a body model calibrated to the measurements; cheap introspection sits below it, full telemetry far above.

The break-even is robust across the model’s body parameters—substrate-conditional rational self-blindness, not a universal limit (Fig. 5).

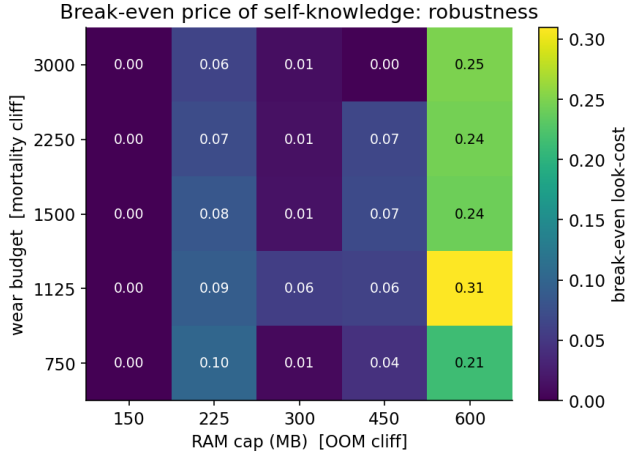


Figure 5: The break-even is robust across the model’s body parameters (RAM cap, wear budget).

5 The self-sensing-availability axis

Substrates differ in whether self-sensing is possible at all: rich in-process (Linux CPU), costly subprocess (GPU), platform-sandboxed (iOS and Android restrict /proc), and absent (a Cloud TPU v5e exposes no readable thermal source). The last are platform-enforced self-blindness.

6 Discussion

The measured back-action is one regime of a general principle: self-measurement perturbs the measured wherever sensor and substrate are shared—digital (energetic and structural cost), analog (the loading effect), quantum (measurement collapse), severity increasing from cost to perturbation to destruction. We measure the digital and quantum regimes and cite the analog as established; this is a unifying lens, not a claim of one mechanism.

7 Implications for machine introspection

If self-report is bounded by a measurement back-action, a model cannot obtain a complete consistent snapshot of its own state—a limit additional to confabulation. This partly explains why measured introspective reliability is low [9], and predicts an irreducible residual that training

cannot remove. It also gives a capable embodied agent a cost-rational incentive against self-transparency, distinct from deception.

8 Limitations

We lead with these. (1) We extend and measure Breuer; we do not re-derive the impossibility. (2) The thermal mechanism is ordinary thermodynamics, the informational floor ordinary self-reference, and quantum back-action textbook; the contribution is the measurement, the dynamic framing, and the three-channel demonstration. (3) The informational floor is structural, not thermodynamic. (4) Thermal hysteresis is shown on one bare-metal CPU plus three GPUs; absolute magnitudes vary with cooling. (5) The quantum penalty is proven in closed form (projective vs. POVM, via Naimark) and confirmed on a real QPU within device noise. (6) Apple-Silicon CPU temperature and virtualized thermal are unmeasured. (7) The introspection implication is an explicit analogy between substrate-level self-measurement and representational self-report.

9 Conclusion

Self-knowledge on real substrates is bounded twice—by cost, and by an irreducible self-perturbation measured in three independent channels that extends a known self-measurement impossibility from static indistinguishability to measured dynamic back-action. A machine cannot cheaply, and cannot fully, read its own physical state; it cannot read its own resting state at all.

Reproducibility. All code, per-substrate results, and figure scripts are available at <https://github.com/dcondrey/cost-of-self-knowledge>: `benchmark.py` (cost), `selfmeasure.py` (thermal), `snapshot.py` (informational), `quantum_theory.py` and `quantum_frontier.py` (quantum), with all `results/*.json`.

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